

Concord grape juice supplementation improves memory function in older adults with mild cognitive impairment.

Concord grape juice contains polyphenol compounds, which have antioxidant and anti-inflammatory properties and influence neuronal signalling. Concord grape juice supplementation has been shown to reduce inflammation, blood pressure and vascular pathology in individuals with CVD, and consumption of such flavonoid-containing foods is associated with a reduced risk for dementia. In addition, preliminary animal data have indicated improvement in memory and motor function with grape juice supplementation, suggesting potential for cognitive benefit in ageing humans. In this initial investigation of neurocognitive effects, we enrolled twelve older adults with memory decline but not dementia in a randomised, placebo-controlled, double-blind trial with Concord grape juice supplementation for 12 weeks. We observed significant improvement in a measure of verbal learning and non-significant enhancement of verbal and spatial recall. There was no appreciable effect of the intervention on depressive symptoms and no effect on weight or waist circumference. A small increase in fasting insulin was observed for those consuming grape juice. These preliminary findings suggest that supplementation with Concord grape juice may enhance cognitive function for older adults with early memory decline and establish a basis for more comprehensive investigations to evaluate potential benefit and assess mechanisms of action.